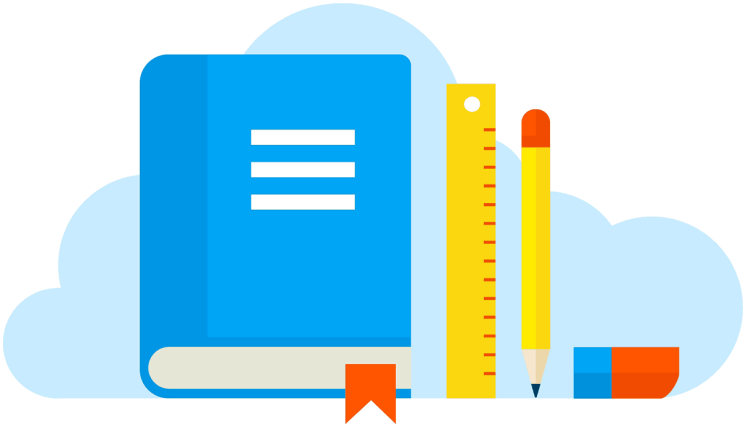


📘 NOTE 1 — HTML & CSS FUNDAMENTALS

Date: 09/04/2026

📘 NOTE 1 — HTML & CSS FUNDAMENTALS

Subject: Web Development
Suggested file name: `HTML_CSS_Fundamentals.pdf`

GOOGLE DOCS FORMATTING

Use:

- **Font: Aptos / Poppins**
- **Main Title: 22 pt, Bold**
- **Headings: 16 pt, Bold**
- **Subheadings: 13–14 pt, Bold**
- **Normal text: 11–12 pt**
- **Line spacing: 1.15**
- **Page: A4**
- **Alignment: Justified**
- **Heading color: Dark blue**
- **Page numbers: Bottom center**

- **Add a thin line below the main title for a professional look.**

HTML & CSS FUNDAMENTALS

Web Development Notes

Subject: Web Development
Topic: HTML & CSS Fundamentals
Level: BCA Semester 5

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1. Introduction to Web Development

Web development is the process of creating websites and web applications that run inside a web browser. A website generally contains three major parts: HTML, CSS, and JavaScript.

- **HTML defines the structure and content of a webpage.**
- **CSS controls the appearance and layout of the webpage.**
- **JavaScript adds interactivity and dynamic behavior.**

For example, HTML can create a button, CSS can change its color and size, and JavaScript can perform an action when the user clicks the button.

---

2. HTML

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages.

HTML uses elements and tags to define different types of content such as headings, paragraphs, images, links, tables, and forms.

Basic HTML Structure

```
<!DOCTYPE html>
<html>
<head>
    <title>My Web Page</title>
</head>

<body>
    <h1>Welcome to StudySphere</h1>
    <p>This is a learning platform.</p>
</body>

</html>
```

Important HTML Tags

`<h1>` to `<h6>`
Used to create headings.

`<p>`
Used to create paragraphs.

`<a>`
Used to create hyperlinks.

```
<a href="https://example.com">Visit Website</a>
```

`<img>`
Used to display an image.

```
<img src="image.jpg" alt="Study Material">
```

`<ul>` and `<ol>`
Used to create unordered and ordered lists.

```
<ul>
    <li>HTML</li>
    <li>CSS</li>
```

```
    <li>JavaScript</li>
</ul>
```

---

3. HTML Forms

HTML forms are used to collect information from users.

Common form elements include:

- **Text box**
- **Email field**
- **Password field**
- **Radio button**
- **Checkbox**
- **Submit button**
- **Dropdown list**

Example

```
<form>
    <label>Name:</label>
    <input type="text">

    <label>Email:</label>
    <input type="email">

    <button type="submit">Register</button>
</form>
```

Forms are commonly used for registration, login, assignment submission, search, and feedback.

---

4. CSS

CSS (Cascading Style Sheets) is used to control the visual appearance of HTML elements.

CSS can control:

- **Colors**
- **Fonts**
- **Text size**
- **Spacing**
- **Borders**
- **Backgrounds**
- **Width and height**
- **Page layout**
- **Responsive design**

Basic CSS Syntax

```
selector {
    property: value;
}
```

Example

```
h1 {
    color: blue;
    font-size: 30px;
}
```

Here:

- `h1` **is the selector**
- `color` **is the property**
- `blue` **is the value**

5. Types of CSS

There are three main ways to apply CSS.

1. Inline CSS

CSS is written directly inside an HTML element.

```
<p style="color: blue;">Welcome to StudySphere</p>
```

2. Internal CSS

CSS is written inside the `<style>` tag in the HTML document.

```
<style>
    p {
        color: blue;
    }
</style>
```

3. External CSS

CSS is written in a separate `.css` file and linked with HTML.

```
<link rel="stylesheet" href="style.css">
```

External CSS is generally preferred for larger websites because it keeps the HTML and styling separate and makes the code easier to maintain.

---

6. CSS Selectors

CSS selectors are used to select HTML elements that need styling.

Element Selector

```
p {
    font-size: 16px;
}
```

It applies the style to all <p> elements.

Class Selector

```
.note {
    background-color: lightgray;
}
```

HTML:

```
<div class="note">Study Material</div>
```

ID Selector

```
#header {
    background-color: navy;
}
```

HTML:

```
<div id="header">StudySphere</div>
```

---

7. CSS Box Model

The CSS Box Model describes how space is calculated around an HTML element.

It consists of:

Content → Padding → Border → Margin

- Content: Actual text or element content.
- Padding: Space between content and border.

- **Border: Line surrounding the element.**
- **Margin: Space outside the border.**

Example:

```
.card {
    padding: 20px;
    border: 1px solid gray;
    margin: 15px;
}
```

The box model is important for creating properly spaced and organized web pages.

---

8. HTML and CSS Together

HTML provides the structure, while CSS provides the design.

Example:

```
<div class="card">
    <h2>StudySphere</h2>
    <p>Student Learning Platform</p>
</div>
```

```
.card {
    width: 300px;
    padding: 20px;
    border-radius: 10px;
    background: white;
}

.card h2 {
    color: #2563eb;
}
```

In this example, HTML creates the card structure and CSS makes the card visually attractive.

---

9. Importance of HTML and CSS

HTML and CSS are fundamental technologies in web development.

HTML is used for:

- **Creating webpage structure**
- **Adding text and images**
- **Creating links**
- **Creating forms**
- **Organizing content**

CSS is used for:

- **Designing webpages**
- **Creating layouts**
- **Applying colors and fonts**
- **Adding spacing and borders**
- **Making websites responsive**

Together, HTML and CSS provide the basic foundation for modern websites and web applications.

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10. Conclusion

HTML and CSS are essential technologies for web development. HTML creates the structure of a webpage, while CSS controls its appearance and layout. Understanding both technologies provides a strong foundation for learning JavaScript, React, and other modern web development frameworks.